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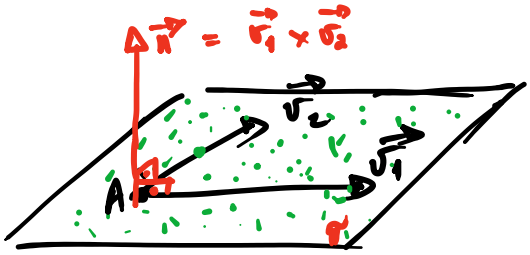
Exemplo

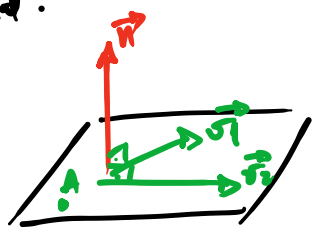
Seja π o plano que passa por A(2,2,-1) e é paralelo aos vetores $\vec{u} = (2, -3, 1)$ e $\vec{v} = (-1, 5, -3)$. Obtenha as equações paramétricas de π e sua equação geral.



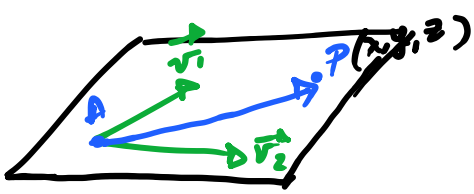
$$\begin{cases} 2x = 2\cdot 2 + 2\cdot t\cdot 2 - 1\cdot h\,(\cdot 2) \\ y = 2 - 3\cdot t + 5\cdot h \\ z = -1 + 1\cdot t - 3\cdot h \end{cases}, \quad t, h \in \mathbb{R}$$

possibilidades p/ eq. geral:

(I) $\vec{n} = \vec{v_1} \times \vec{v_2}$



(II)



$\vec{v_1}, \vec{v_2}$ e $\vec{AP}$ são coplanares

$(\vec{AP}, \vec{v_1}, \vec{v_2}) = 0.$

(I) $\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} 1 & 1 & k \\ 2 & -3 & 1 \\ -1 & 5 & -3 \end{vmatrix} \begin{matrix} 1 & 1 \\ 2 & -3 \\ -1 & 5 \end{matrix}$

$-3\vec{k} \quad -5\vec{i} \quad 6\vec{j} \quad 9\vec{i} \quad -\vec{j} \quad 10\vec{k}$

$\vec{n} = 4\vec{i} + 5\vec{j} + 7\vec{k} = (4,5,7)$

$\pi: \quad ax + by + cz + d = 0$

$4x + 5y + 7z + d = 0$

$A = (2,2,-1) \in \pi \Rightarrow$

$4 \cdot 2 + 5 \cdot 2 + 7 \cdot (-1) + d = 0$

$8 + 10 - 7 + d = 0$

$11 + d = 0 \Rightarrow \underline{d = -11}$

$$\boxed{\pi: 4x + 5y + 7z - 11 = 0}$$

Verifique!!!

(II) $\vec{AP} = P - A = (x,y,z) - (2,2,-1)$

$\vec{AP} = (x-2, y-2, z+1)$

$$(\vec{AP}, \vec{v}_1, \vec{v}_2) = \begin{vmatrix} x-2 & y-2 & z+1 \\ 2 & -3 & 1 \\ -1 & 5 & -3 \end{vmatrix} = 0$$

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Exercício

Determinar o valor de m para que seja $30°$ o ângulo entre os planos

$$\pi_1 : x + my + 2z - 7 = 0 \qquad \text{e} \qquad \pi_2 : 4x + 5y + 3z + 2 = 0$$

$\vec{n_1} = (1, m, 2)$, $\vec{n_2} = (4, 5, 3)$

ângulo entre π_1 e π_2 = ângulo entre $\vec{n_1}$ e $\vec{n_2}$

$$\cos(\theta) = \frac{|\vec{n_1} \cdot \vec{n_2}|}{|\vec{n_1}| \cdot |\vec{n_2}|}$$

$$\cos(30) = \frac{|(1, m, 2) \cdot (4, 5, 3)|}{\sqrt{1^2 + m^2 + 2^2} \cdot \sqrt{4^2 + 5^2 + 3^2}}$$

$$\frac{\sqrt{3}}{2} = \frac{|4 + 5m + 6|}{\sqrt{1 + m^2 + 4} \cdot \sqrt{16 + 25 + 9}}$$

$$\frac{\sqrt{3}}{2} = \frac{|5m + 10|}{\sqrt{m^2 + 5} \cdot \sqrt{50}} \quad , \quad \sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$$

$$\left(\frac{\sqrt{3}}{2}\right)^2 = \left(\frac{\cancel{5}|m + 2|}{\sqrt{m^2 + 5} \cdot \cancel{5} \cdot \sqrt{2}}\right)^2$$

$$\frac{3}{4} = \frac{(m + 2)^2}{2m^2 + 10}$$